EXPLORATORY DATA ANALYSIS (EDA) ON RETAIL SALES DATA

Project Overview

This project focuses on performing Exploratory Data Analysis (EDA) on retail sales data to identify patterns, trends, and insights. The dataset includes transaction details such as customer information, product categories, and sales amounts. The goal is to analyze the data and provide meaningful insights to support business decision-making.

Project Objectives

- To analyze retail sales data
- To identify trends and patterns in sales
- To understand customer behavior
- To visualize data using charts and graphs
- To provide actionable business recommendations

Dataset Overview

The dataset contains 1000 retail transactions with 9 columns.

Tools & Technologies Used

- **Python**
- **Pandas**
- **NumPy**
- **Matplotlib**
- **Seaborn**
- **Jupyter Notebook**

Data Cleaning & Preprocessing

- **No missing values were found in the dataset**
- **Data types were verified and corrected**
- **Date column was converted to datetime format**
- **New features (Year, Month) were created for analysis**

Exploratory Data Analysis

Monthly Sales Trend (Line Plot)

Insight: Sales peaked in May (Month 5) at ~₹52,000, showing a clear seasonal high. September (Month 9) recorded the lowest sales at ~₹24,000. A recovery was observed in October and December. The irregular pattern suggests demand is driven by seasonal factors or planned promotions.

Recommendation: Prepare inventory and staffing in advance for peak months like May. Run special discounts or loyalty offers during slow months like September to boost sales.

Sales by Product Category (Bar Chart)

Insight: All three categories performed similarly — Electronics (₹1,56,905) > Clothing (₹1,55,580) > Beauty (₹1,43,515). No single category dominates, meaning the business has a well-balanced product portfolio.

Recommendation: Beauty category is slightly behind — targeted promotions or bundle offers can help close the gap. Upselling accessories and warranties in Electronics can further increase revenue.

Sales by Gender (Bar Chart)

Insight: Female customers contributed ₹2,32,840 while Male customers contributed ₹2,23,160 — a difference of

only ~4%. Both genders contribute almost equally to total sales.

Recommendation: Gender-neutral marketing campaigns will be effective. However, a product-level gender breakdown could reveal more specific buying patterns worth targeting.

Sales by Age (Bar Chart)

Insight: Sales are not evenly distributed across age groups. Notable peaks are at Age 43 (₹17,970), Age 34 (₹16,785), and Age 51 (₹16,065). Younger customers aged 18–25 tend to spend comparatively less.

Recommendation: The 30–55 age group is the most valuable customer segment — focus marketing and premium products on this group. To attract younger customers (18–25), consider introducing budget-friendly ranges or EMI/installment options.

Top 5 Customers (Bar Chart)

Insight: The top 5 customers (CUST487, CUST476, CUST773, CUST503, CUST093) each spent exactly ₹2,000 — the maximum transaction value in the dataset. These are likely single high-value purchases, possibly in Electronics.

Recommendation: Introduce a VIP loyalty program with exclusive discounts for high-spending customers. Retaining existing high-value customers is more cost-effective than acquiring new ones.

Correlation Heatmap

Insight: **The strongest correlation is between Price per Unit and Total Amount (0.85) — meaning higher-priced products directly drive higher revenue. Quantity and Total Amount also show a moderate correlation (0.37). Age has no meaningful correlation with any variable, meaning age alone cannot predict purchasing behavior.**

Recommendation: **Promoting premium/high-price products is the most effective strategy to increase revenue. Bundle deals that increase quantity per transaction can also meaningfully boost total sales.**

Final Overall Recommendations

1. Seasonal Strategy: May is the peak month — plan stock and campaigns in advance. Run flash sales in September to counter the seasonal dip.

2. Product Focus: All three categories are balanced, but Beauty needs a slight push through targeted offers or combo deals.

3. Customer Segmentation: Target the 30–55 age group with premium products. Introduce budget-friendly options for younger customers aged 18–25.

4. VIP Program: Launch a loyalty rewards program for top spenders to improve retention.

5. Pricing Strategy: High-price items are the strongest driver of revenue — consider expanding the premium product line.

Retail_Sales_EDA_Project

localhost:8888/notebooks/Retail_Sales_EDA_Project.ipynb

jupyter Retail_Sales_EDA_Project Last Checkpoint: 2 minutes ago

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Code

JupyterLab Python 3 (ipykernel)

[1]:
```
import pandas as pd

df = pd.read_csv("retail_sales_dataset.csv")
df.head()
```

[1]:

	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
0	1	2023-11-24	CUST001	Male	34	Beauty	3	50	150
1	2	2023-02-27	CUST002	Female	26	Clothing	2	500	1000
2	3	2023-01-13	CUST003	Male	50	Electronics	1	30	30
3	4	2023-05-21	CUST004	Male	37	Clothing	1	500	500
4	5	2023-05-06	CUST005	Male	30	Beauty	2	50	100

The dataset contains retail transaction data including customer details, product categories, and sales information.

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Code

JupyterLab Python 3 (ipykernel)

[2]:
```
df.shape
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Transaction ID    1000 non-null   int64
 1   Date              1000 non-null   object
 2   Customer ID       1000 non-null   object
 3   Gender            1000 non-null   object
 4   Age               1000 non-null   int64
 5   Product Category  1000 non-null   object
 6   Quantity          1000 non-null   int64
 7   Price per Unit    1000 non-null   int64
 8   Total Amount      1000 non-null   int64
dtypes: int64(5), object(4)
memory usage: 70.4+ KB
```

[2]:

	Transaction ID	Age	Quantity	Price per Unit	Total Amount
count	1000.000000	1000.00000	1000.000000	1000.000000	1000.000000
mean	500.500000	41.39200	2.514000	179.890000	456.000000
std	288.819436	13.68143	1.132734	189.681356	559.997632
min	1.000000	18.00000	1.000000	25.000000	25.000000
25%	250.750000	29.00000	1.000000	30.000000	60.000000
50%	500.500000	42.00000	3.000000	50.000000	135.000000
75%	750.250000	53.00000	4.000000	300.000000	900.000000
max	1000.000000	64.00000	4.000000	500.000000	2000.000000

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Code

JupyterLab Python 3 (ipykernel)

mean	500.500000	41.39200	2.514000	179.890000	456.000000
std	288.819436	13.68143	1.132734	189.681356	559.997632
min	1.000000	18.00000	1.000000	25.000000	25.000000
25%	250.750000	29.00000	1.000000	30.000000	60.000000
50%	500.500000	42.00000	3.000000	50.000000	135.000000
75%	750.250000	53.00000	4.000000	300.000000	900.000000
max	1000.000000	64.00000	4.000000	500.000000	2000.000000

Insight: The dataset contains 1000 rows and 9 columns There are no missing values in the dataset The average sales amount is around 456 Customer age ranges from 18 to 64

[3]:
```
df.isnull().sum()
```

[3]:
```
Transaction ID      0
Date                0
Customer ID         0
Gender              0
Age                 0
Product Category    0
Quantity            0
Price per Unit      0
Total Amount        0
dtype: int64
```

No missing values are present in the dataset

[4]:
```
df.dropna(inplace=True)
```

Since there were no missing values, no data cleaning was required.

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Code

JupyterLab Python 3 (ipykernel)

[5]:
```
df.describe()
```

[5]:

	Transaction ID	Age	Quantity	Price per Unit	Total Amount
count	1000.000000	1000.00000	1000.000000	1000.000000	1000.000000
mean	500.500000	41.39200	2.514000	179.890000	456.000000
std	288.819436	13.68143	1.132734	189.681356	559.997632
min	1.000000	18.00000	1.000000	25.000000	25.000000
25%	250.750000	29.00000	1.000000	30.000000	60.000000
50%	500.500000	42.00000	3.000000	50.000000	135.000000
75%	750.250000	53.00000	4.000000	300.000000	900.000000
max	1000.000000	64.00000	4.000000	500.000000	2000.000000

Insight: The average customer age is ~41 years (range: 18–64). The average transaction amount is ₹456, but the median is only ₹135 — this means a few high-value transactions (max ₹2000) are pulling the average up. Quantity always stays between 1–4. Price per unit varies widely (₹25 to ₹500), reflecting different product categories.

[8]:
```
df['Date'] = pd.to_datetime(df['Date'])
```

[9]:
```
df['Year'] = df['Date'].dt.year
df['Month'] = df['Date'].dt.month
```

Date column was converted into datetime format New features like Year and Month were extracted for time analysis

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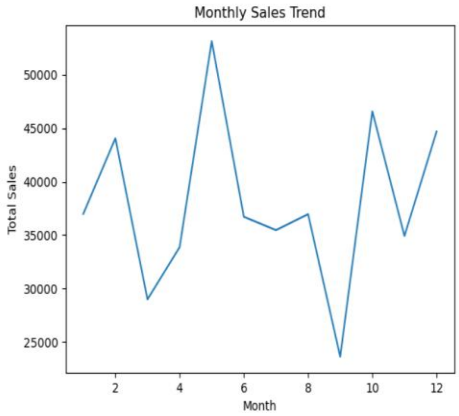
Code

JupyterLab Python 3 (ipykernel)

[10]:
```
import matplotlib.pyplot as plt

df.groupby('Month')['Total Amount'].sum().plot()
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Total Sales")
plt.show()
```



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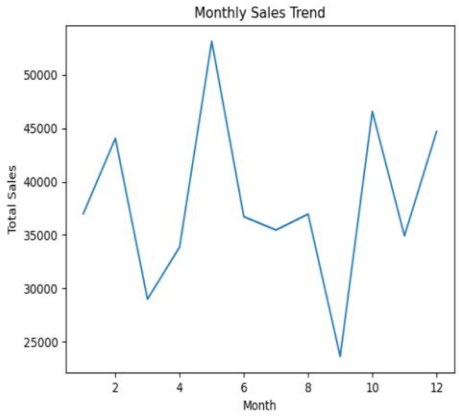
Code

JupyterLab Python 3 (ipykernel)

```
plt.show()
```



Insight: Sales peaked in May (Month 5) at ~₹52,000, showing a clear seasonal high. September (Month 9) recorded the lowest sales at ~₹24,000. A recovery was observed in October and December. The irregular pattern suggests demand is driven by seasonal factors or planned promotions.

Recommendation: Prepare inventory and staffing in advance for peak months like May. Run special discounts or loyalty offers during slow months like September to boost sales.

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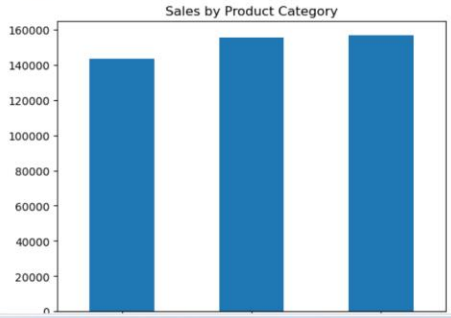
Code

JupyterLab Python 3 (ipykernel)

[11]:
```
df.groupby('Product Category')['Total Amount'].sum().sort_values(ascending=False)
```

[11]:
```
Product Category
Electronics    156905
Clothing       155580
Beauty         143515
Name: Total Amount, dtype: int64
```

[12]:
```
df.groupby('Product Category')['Total Amount'].sum().plot(kind='bar')
plt.title("Sales by Product Category")
plt.show()
```



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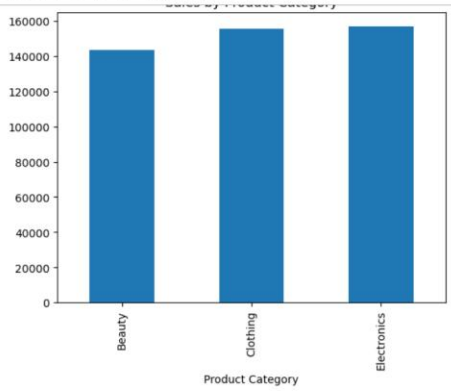
Code

JupyterLab Python 3 (ipykernel)



Insight: All three categories performed similarly — Electronics (₹1,56,905) > Clothing (₹1,55,580) > Beauty (₹1,43,515). No single category dominates, meaning the business has a well-balanced product portfolio.

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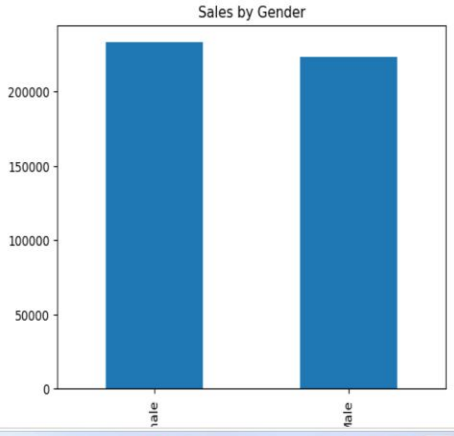
Code

JupyterLab Python 3 (ipykernel)

[13]:
```
df.groupby('Gender')['Total Amount'].sum()
```

[13]:
```
Gender
Female    232840
Male      223160
Name: Total Amount, dtype: int64
```

[14]:
```
df.groupby('Gender')['Total Amount'].sum().plot(kind='bar')
plt.title("Sales by Gender")
plt.show()
```



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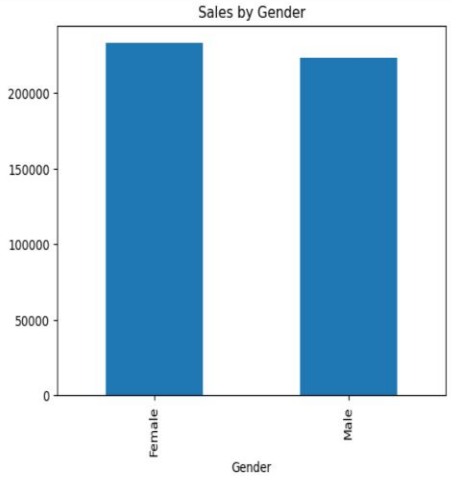
Code

JupyterLab Python 3 (ipykernel)



Insight: Female customers contributed ₹2,32,840 while Male customers contributed ₹2,23,160 — a difference of only ~4%. Both genders contribute almost equally to total sales.

Recommendation: Gender-neutral marketing campaigns will be effective. However, a product-level gender breakdown could reveal more specific buying patterns worth targeting.

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Code

JupyterLab Python 3 (ipykernel)

[15]:
```
df.groupby('Age')['Total Amount'].sum()
```

[15]:
```
Age
18    11215
19    14870
20     8645
21    12585
22    13700
23     8220
24     5415
25     9900
26    13980
27     9385
28     8670
29     6570
30     9790
31    10220
32     5550
33     6240
34    16785
35    11290
36     9105
37    11650
38    11100
39     4595
40     9415
41     5650
42     8500
43    17970
44     7560
45     6325
46    13090
47    12505
48     7240
49     5110
50     9845
```

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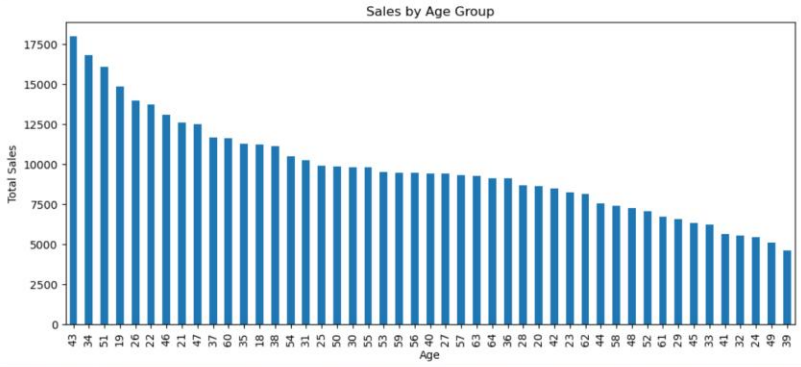
Code

JupyterLab Python 3 (ipykernel)

[17]:
```
age_sales = df.groupby('Age')['Total Amount'].sum()

age_sales.sort_values(ascending=False).plot(kind='bar', figsize=(12,5))
plt.title("Sales by Age Group")
plt.xlabel("Age")
plt.ylabel("Total Sales")
plt.show()
```



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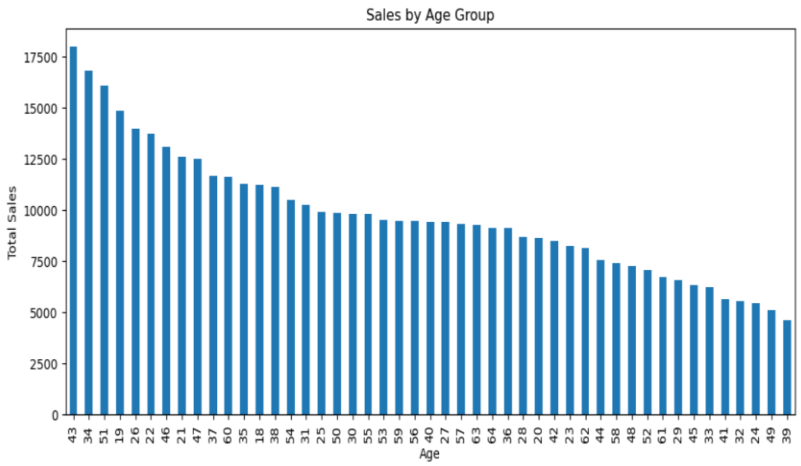
Code

JupyterLab Python 3 (ipykernel)



Insight: Sales are not evenly distributed across age groups. Notable peaks are at Age 43 (₹17,970), Age 34 (₹16,785), and Age 51 (₹16,065). Younger customers aged 18–25 tend to spend comparatively less.

Recommendation: The 30–55 age group is the most valuable customer segment — focus marketing and premium products on this group. To attract younger customers (18–25), consider introducing budget-friendly ranges or EMI/installment options.

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Code

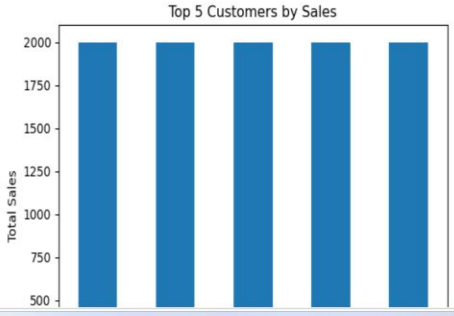
JupyterLab Python 3 (ipykernel)

[18]:
```
df.groupby('Customer ID')['Total Amount'].sum().sort_values(ascending=False).head()
```

[18]:
```
Customer ID
CUST487    2000
CUST476    2000
CUST773    2000
CUST503    2000
CUST093    2000
Name: Total Amount, dtype: int64
```

[20]:
```
top_customers = df.groupby('Customer ID')['Total Amount'].sum().sort_values(ascending=False).head()

top_customers.plot(kind='bar')
plt.title("Top 5 Customers by Sales")
plt.xlabel("Customer ID")
plt.ylabel("Total Sales")
plt.show()
```



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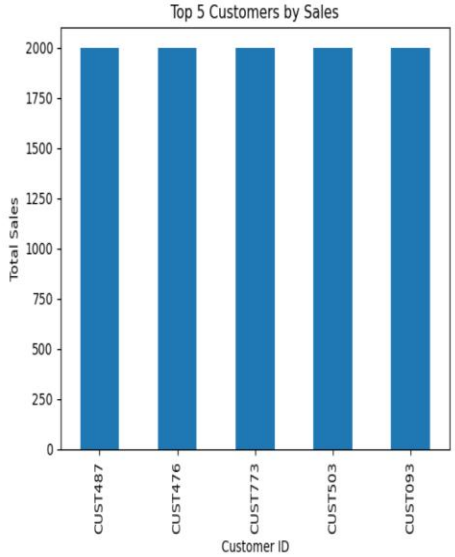
Code

JupyterLab Python 3 (ipykernel)



Insight: The top 5 customers (CUST487, CUST476, CUST773, CUST503, CUST093) each spent exactly ₹2,000 — the maximum transaction value in the dataset. These are likely single high-value purchases, possibly in Electronics.

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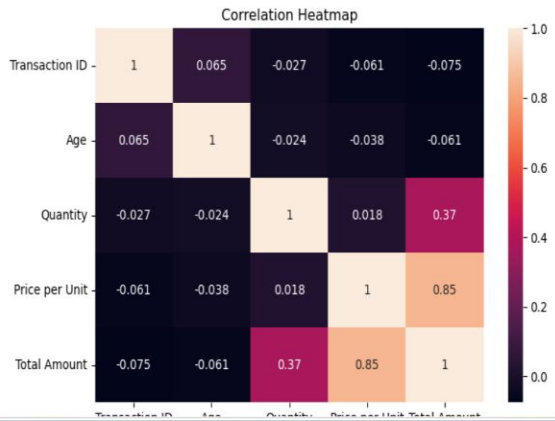
Code

JupyterLab Python 3 (ipykernel)

[22]:
```
import seaborn as sns
import matplotlib.pyplot as plt

numeric_df = df.select_dtypes(include=['int64', 'float64'])

plt.figure(figsize=(8,5))
sns.heatmap(numeric_df.corr(), annot=True)
plt.title("Correlation Heatmap")
plt.show()
```



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Retail_Sales_EDA_Project
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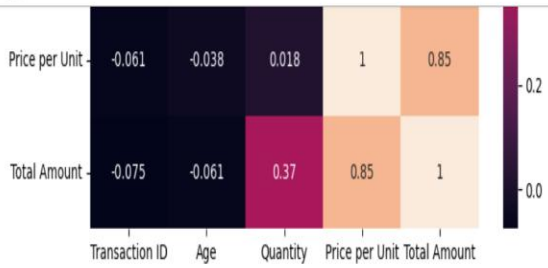
Code
JupyterLab
Python 3 (ipykernel)



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Recommendation: Promoting premium/high-price products is the most effective strategy to increase revenue. Bundle deals that increase quantity per transaction can also meaningfully boost total sales

Final Overall Recommendations :

1. Seasonal Strategy: May is the peak month — plan stock and campaigns in advance. Run flash sales in September to counter the seasonal dip.
2. Product Focus: All three categories are balanced, but Beauty needs a slight push through targeted offers or combo deals.
3. Customer Segmentation: Target the 30–55 age group with premium products. Introduce budget-friendly options for younger customers aged 18–25.
4. VIP Program: Launch a loyalty rewards program for top spenders to improve retention.
5. Pricing Strategy: High-price items are the strongest driver of revenue — consider expanding the premium product line.

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